

# The allegory axioms: the proofs

The companion to `allegory-axioms.typ`, which states these laws and their pictures. Here each one is worked: the steps of the Lean proof, side by side, with the rule that reaches each under it.

## §1. $R \cap R = R$ , the one that is not bookkeeping

$$\boxed{R} = \boxed{R} \begin{array}{c} \diagup \\ \diagdown \end{array} = \boxed{R} \begin{array}{c} \diagup \\ \diagdown \end{array} \leq \begin{array}{c} \boxed{R} \\ \boxed{R} \end{array} \begin{array}{c} \diagup \\ \diagdown \end{array} \text{ lax copy}$$

$\triangleleft \triangleright = 1$

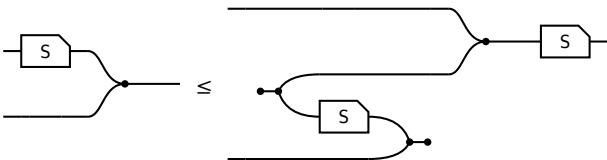
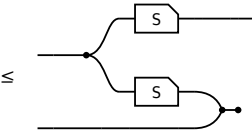
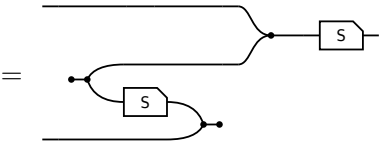
The last picture is  $R \cap R$  itself, so this is  $R \leq R \cap R$  and the lax copy law is the whole of it. The other direction is not proved here:  $R \cap S \leq R$  holds for every  $S$ , by weakening  $S$  to  $\top$  and then collapsing  $R \cap \top$  — which is the paper's own remaining three steps, packaged once.

## §2. The modular law, from Frobenius

| <p>the modular law, <math>R S \cap T \leq (R \cap T S^\circ) S</math></p> |         |                                                                                                                                 |
|---------------------------------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------|
| term                                                                      | picture | the rule that reaches it                                                                                                        |
| $R S \cap T$                                                              |         | the start: $R S \cap T$ .                                                                                                       |
| $= \triangleleft (R \otimes T) (S \otimes 1) \triangleright$              |         | Factor out the shared prefix: $(R S) \otimes T$ is $(R \otimes T) (S \otimes 1)$ .                                              |
| $\leq \triangleleft (R \otimes T) (1 \otimes S^\circ) \triangleright S$   |         | The merge slide. $S$ slides through the merge and reappears as $S^\circ$ on the other strand. The only inequality in the proof. |
| $= (R \cap T S^\circ) S$                                                  |         | Fold the prefix back in; the copy-merge pair reads as a meet again.                                                             |

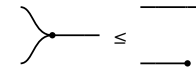
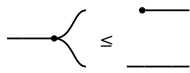
$S$  occurs once on the left of the slide and twice on the right, and the lax copy law is the only law in the definition that may duplicate a box — so that is where the inequality has to be.

| <p>the merge slide, <math>(S \otimes 1) \triangleright \leq (1 \otimes S^\circ) \triangleright S</math></p> |         |                                                                                           |
|-------------------------------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------|
| term                                                                                                        | picture | the rule that reaches it                                                                  |
| $(S \otimes 1) \triangleright$                                                                              |         | the start: $(S \otimes 1) \triangleright$ .                                               |
| $= ((S \triangleleft) \otimes 1) (1 \otimes \triangleright \circ)$                                          |         | Rebuild the merge from a copy and a right bracket, the shape the lax copy law applies to. |

| <p>the merge slide, <math>(S \otimes 1) \triangleright \leq (1 \otimes S^\circ) \triangleright S</math></p>  |                                                                                    |                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| term                                                                                                                                                                                           | picture                                                                            | the rule that reaches it                                                                                                                      |
| $\leq ((\triangleleft (S \otimes S)) \otimes 1)(1 \otimes \triangleright \circ)$                                                                                                               |  | <b>The lax copy law:</b> $S \triangleleft$ becomes $\triangleleft (S \otimes S)$ , the box copied. The whole of the modular law, in one step. |
| $= (1 \otimes S^\circ) \triangleright S$                                                                                                                                                       |  | <b>Reshape back</b> , and the surviving duplicate slides round the right bracket to become $S^\circ$ .                                        |

### §3. $R \leq R R^\circ R$ , by cutting two wires

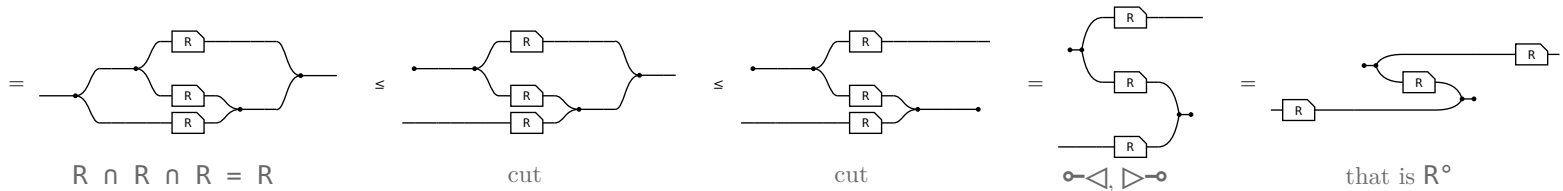
Freyd reads this off the modular law (§2.112):  $R \sqsubseteq 1R \sqcap R \sqsubseteq (1 \sqcap R R^\circ) R \sqsubseteq R R^\circ R$ . In pictures neither the modular law nor a meet with 1 is wanted. The one law that makes a picture bigger is spent twice, and it is the cut,  $1 \leq \circ \triangleleft$ :



cut the copy's left output and what comes out there is **anything at all**

cut the merge's right input and it is **simply discarded**

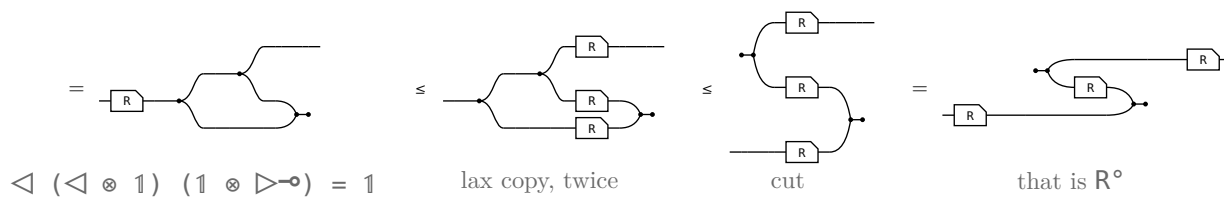
$R$  is its own three-fold meet — three copies of the box between a copy tree and a merge tree — and the two cuts turn the trees into a  $\circ \triangleleft$  and a  $\triangleright \circ$ . The three boxes never move:



Read the last picture from the input, which is now the bottom strand: through the bottom box, up the  $\triangleright \circ$  and back along the middle box — which is that box turned round, so it is  $R^\circ$  — down the  $\circ \triangleleft$  and out along the top box. Where the meet had one input feeding all three boxes and one output draining all three, the zig-zag has both of those **cut**, and a cut wire is the one thing this calculus lets you add for free.

### §4. The same, from the cap end

The section above cuts twice in its chain. This one cuts once and pays for the other cut with the **lax copy law** —  $R \triangleleft \leq \triangleleft (R \otimes R)$ , the only law in the definition that may duplicate a box. Read 1 as a copy tree whose last two legs are capped back:



The three boxes are not there to begin with: there is one, and the two copy dots walk back **through** it, leaving it behind three times. So the  $\triangleright \circ$  is the frame's own cap before anything is cut, and the copy tree is the only thing left to cut.

Neither chain is cheaper in what it **spends**.  $\triangleleft \triangleright = 1$  is itself the merge cut, so the second  $1 \leq \circ \triangleleft$  is underneath this one;  $R \sqcap R = R$  is the lax copy law, so that is underneath the section above. Both routes pay both. What differs is the **layer**: this chain names only the copy, the cap and the box, so it holds as soon as the converse does — before  $\sqcap$  exists. The section above climbs into the meet semilattice and back down to state a containment in which no meet occurs.

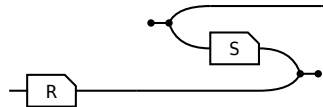
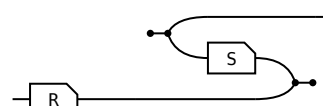
### §5. Entireness of an intersection

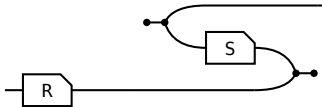
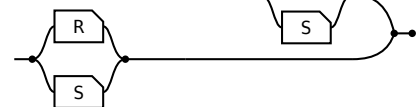
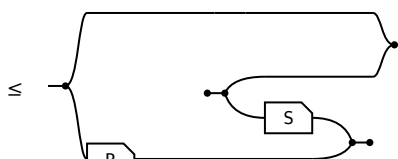
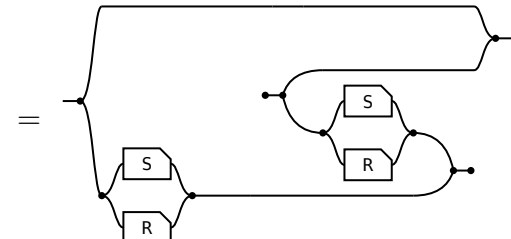
$R$  is **entire** when  $1 \sqsubseteq R R^\circ$ . When is an **intersection** entire?



$$\text{Total } (R \cap S) \leftrightarrow 1 \leq R S^\circ$$

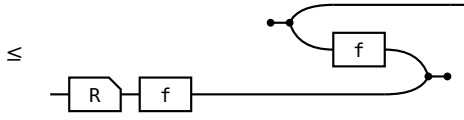
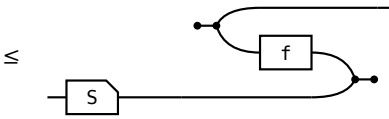
On the left  $R \cap S$  is named **twice**; on the right  $R$  and  $S$  once each and the meet is gone. The meet is needed to **ask** whether something is entire, not to **answer** it. Where  $\sqsubseteq$  is **defined** as  $X \cap Y = X$  the two sides are one proposition; here  $\leq$  is primitive, so both directions are real steps — and they cost very different things.

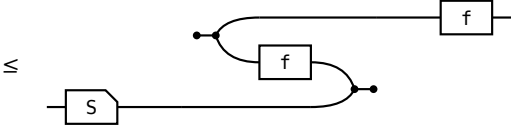
| $\Rightarrow$ given $R \cap S$ entire,  show  |                                                                                      |                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------|
| term                                                                                                                                                                                                               | picture                                                                              | the rule that reaches it                               |
| 1                                                                                                                                                                                                                  |     | the start: 1.                                          |
| $\leq (R \cap S) (R \cap S)^\circ$                                                                                                                                                                                 |   | the hypothesis: $1 \leq P P^\circ$ at $P = R \cap S$ . |
| $\leq R S^\circ$                                                                                                                                                                                                   |  | monotonicity, twice. No modular law.                   |

| $\Leftarrow$ given  show $R \cap S$ entire,  |                                                                                      |                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| term                                                                                                                                                                                                                 | picture                                                                              | the rule that reaches it                                                                                                                                                                                                                                                                                    |
| 1                                                                                                                                                                                                                    |   | the start: 1.                                                                                                                                                                                                                                                                                               |
| $\leq 1 \cap R S^\circ$                                                                                                                                                                                              |  | the hypothesis, met with 1.                                                                                                                                                                                                                                                                                 |
| $= 1 \cap (S \cap R) (S \cap R)^\circ$                                                                                                                                                                               |  | $1 \cap R S^\circ = 1 \cap (S \cap R) (S \cap R)^\circ$ . $R S^\circ$ relates $a$ to $a'$ when some $b$ has $a R b$ and $a' S b$ ; the $1 \cap$ sets $a' = a$ , and it reads “some $b$ has $a R b$ and $a S b$ ” — one arrow of $R \cap S$ . Two arrows become one, and that is what needs the modular law. |
| $\leq (R \cap S) (R \cap S)^\circ$                                                                                                                                                                                   |  | $X \cap Y \leq Y$ , then $S \cap R = R \cap S$ .                                                                                                                                                                                                                                                            |

## §6. The shunting rule

A map moves from one side of a containment to the other at the cost of its converse,  $R \circ f \leq S \iff R \leq S \circ f^\circ$ . It is how a function gets out of the way so the relation underneath can be reasoned about. Each direction spends exactly one half of **map** — never both.

| <b>shunting right, <math>\Rightarrow</math></b> given  $\leq$  show  $\leq$  |                                                                                    |                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| term                                                                                                                                                                                                                                                                                                                                                                                                               | picture                                                                            | the rule that reaches it                                                                  |
| $R$                                                                                                                                                                                                                                                                                                                                                                                                                |   | <b>the start:</b> $R$ .                                                                   |
| $\leq R \circ f \circ f^\circ$                                                                                                                                                                                                                                                                                                                                                                                     |  | <b>total:</b> $1 \leq f \circ f^\circ$ , so $f \circ f^\circ$ may be inserted after $R$ . |
| $\leq S \circ f^\circ$                                                                                                                                                                                                                                                                                                                                                                                             |  | Re-bracket to $(R \circ f) \circ f^\circ$ and apply the hypothesis.                       |

| <b>shunting right, <math>\Leftarrow</math></b> given  $\leq$  show  $\leq$  |                                                                                      |                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| term                                                                                                                                                                                                                                                                                                                                                                                                                      | picture                                                                              | the rule that reaches it                                              |
| $R \circ f$                                                                                                                                                                                                                                                                                                                                                                                                               |   | <b>the start:</b> $R \circ f$ .                                       |
| $\leq S \circ f^\circ \circ f$                                                                                                                                                                                                                                                                                                                                                                                            |  | Apply the hypothesis on the left; the two $f$ s are now adjacent.     |
| $\leq S$                                                                                                                                                                                                                                                                                                                                                                                                                  |   | <b>single valued:</b> $f^\circ \circ f \leq 1$ , so the pair cancels. |

Shunting left is the mirror, same shape, map on the other side throughout. What the rule is, is an adjunction:  $f \dashv f^\circ$ , with **total** the unit and **single valued** the counit, so “ $f$  is a map” and “ $f$  has a right adjoint, namely  $f^\circ$ ” are one statement.